

SCHOOL OF COMPUTER SCIENCE AND APPLICATIONS

DEPARTMENT OF SCIENCE AND COMPUTER SCIENCE

TY B.Sc.(IT)

Semester V

AY 2026-27

Course Code: 2506MJET504A

Course Name: Lab course on Data Analytics Tools

Name of the Student:

Roll Number: _____

Course Objectives

1. To introduce students to the fundamentals of Business Intelligence, Data Analytics, and the Power BI environment.
2. To enable students to connect, import, and clean data from multiple heterogeneous sources.
3. To develop skills in transforming and modelling data for effective analysis.
4. To provide hands-on experience in designing charts, dashboards, and interactive reports.
5. To train students in creating calculated fields, measures, and KPIs using DAX.
6. To build students' capability to publish, share, and integrate Power BI reports with other tools such as Excel and Tableau.

Course Outcomes

1. Understand the Power BI interface, architecture, and data connectivity options.
2. Import, clean, and transform data from multiple sources using Power Query.
3. Create data models and establish relationships between tables.
4. Design effective charts and visualizations for business insights.
5. Apply filters, slicers, calculated fields, measures, and KPIs to build interactive reports.
6. Design, publish, and share dashboards, and integrate Power BI with Excel and other BI tools.

Introduction

1. About the workbook:

This workbook is intended to be used by T.Y.B.Sc. (Information Technology) students for the Business Intelligence and Data Analytics Practical Assignments in Semester-V. This workbook is designed by considering all the practical concepts / topics mentioned in the syllabus.

2. The objectives of this workbook are:

- Defining the scope of the course.
- To have continuous assessment of the course and students.
- Providing ready reference for the students during practical implementation.

- Provide more options to students so that they can have good practice before facing the examination.
- Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

3. How to use this workbook:

Each practical of all sections has two sets – SET A and SET B. It is mandatory for students to complete both sets in lab and to be evaluated by subject teachers.

4. Instructions to the students

- Please read the following instructions carefully and follow them during practical. Students are expected to carry this workbook every time they come to the lab for computer practical.
- Students should prepare for the assignment by reading the relevant material which is mentioned in ready reference.
- Instructor will specify which problems to solve in the lab during the allotted slot and student should complete them and get verified by the instructor. However, student should spend additional hours in Lab and at home to cover all workbook assignments if needed.

Students will be assessed for each assignment on a scale from 0 to 5.

Marks	Description
0	Not done
1	Incomplete
2	Late Complete
3	Needs improvement
4	Complete
5	Well Done

5. Instructions to the Instructors

- Make sure that students should follow above instructions.
- Explain the practical and related concepts in around ten minutes using whiteboard if required or by demonstrating the software.
- Evaluate each assignment carried out by a student on a scale of 5 as specified.
- The value should also be entered on the assignment completion sheet of the respective section.

6. Instructions to the Lab Administrator

You have to ensure appropriate hardware and software is made available to each student.

- Operating System – Windows 10 / 11
- Software – Power BI Desktop (latest version), Microsoft Excel
- Internet connectivity for Power BI Service (publishing/sharing)

Editorial Board

MIT Arts, Commerce & Science College, Alandi(D), Pune

Prepared By:

Dr.. Santosh Pandure

Assistant Professor, Department of Computer Science

MIT Arts Commerce & Science College, Alandi(D), Pune.

Assignment Completion Sheet

Sr. No	Assignment	Marks (out of 5)	Teacher Sign
1	Introduction to Power BI Interface and Data Connectivity		
2	Importing and Cleaning Data from Multiple Data Sources		
3	Data Transformation using Power Query		
4	Creating Data Models and Establishing Relationships		
5	Designing Basic Charts for Data Visualization		
6	Using Filters, Slicers, and Parameters to Enhance Interactivity		
7	Creating Calculated Fields, Measures, and KPIs		
8	Designing Dashboards for Business Insights		
9	Publishing and Sharing Dashboards on Power BI Service		
10	Integrating Power BI with Excel		
11	Mini Project — Interactive Dashboard on a Real-World Dataset		
Total			
Total (out of 05)			

Signature of In-charge

Signature of HOD

Signature of Internal Examiner

Signature of External Examiner

Practical 1: Introduction to Power BI Interface and Data Connectivity

Objective: Explore the Power BI Desktop interface, understand its core components, and connect to sample datasets from different sources to load them into the workspace.

SET A

Q1. List and describe the five main components of the Power BI Desktop interface.

Solution:

1. Ribbon (top): Home, Insert, Modeling, View tabs with commands for data, visuals and formatting.
2. Report View (centre canvas): the working area where visuals are placed and arranged on report pages.
3. Fields Pane (right): lists all tables and columns loaded into the model; used to drag fields onto visuals.
4. Visualizations Pane (right, next to Fields): gallery of chart types and formatting controls for the selected visual.
5. Data View and Model View (left-side icons): Data View shows the loaded tables as rows/columns; Model View shows tables and their relationships as a diagram.

Q2. Connect to an Excel workbook containing an Employee table and load it into Power BI.

Sample Data:

EmpID	Name	Dept	Salary
101	Anita Sharma	CS	55000
102	Rahul Verma	IT	48000
103	Priya Nair	CS	52000
104	Sameer Khan	ECE	46000
105	Divya Iyer	IT	51000

Solution:

1. Home tab -> Get Data -> Excel Workbook.
2. Browse and select the .xlsx file, click Open.
3. In the Navigator window, tick the sheet/table 'Employee'.
4. Click Load (or Transform Data first to clean, then Close & Apply).
5. Confirm the table appears in the Fields pane on the right.

Q3. Connect to a CSV file of Sales data and explain how Power BI auto-detects the delimiter.

Sample Data:

OrderID	Product	Qty	Amount
1	Laptop	2	90000
2	Mouse	5	1500
3	Keyboard	3	2400

Note:

Sample file content (sales.csv):

OrderID,Product,Qty,Amount

1,Laptop,2,90000

2,Mouse,5,1500

3,Keyboard,3,2400

Solution:

1. Home tab -> Get Data -> Text/CSV.
2. Select sales.csv and click Open.
3. The preview window shows a 'Delimiter' dropdown; Power BI scans the first rows and pre-selects 'Comma' because commas separate the values consistently.
4. Verify the column headers are detected correctly, then click Load.

Q4. After loading the Employee table (Q2), explore and note the difference between Report View, Data View, and Model View.

Solution:

1. Report View: empty canvas where you can drag EmpID, Name, Dept, Salary onto a Table visual to see them rendered as a report.
2. Data View: shows the raw Employee table exactly as loaded, row by row, with a column-level 'Data type' and formula bar for calculated columns.
3. Model View: shows the Employee table as a single box (since no relationships exist yet) listing its four columns.

Q5. Connect to a public web page table (e.g. a country population table) and load only the required table into Power BI.

Solution:

1. Home tab -> Get Data -> Web.
2. Enter the URL of the page containing the table and click OK.
3. In the Navigator, Power BI lists every HTML table it found on the page; preview each and tick the one matching Country/Population/Region.
4. Click Load.

SET B

Q6. Use 'Enter Data' to manually create a small table of 5 rows and 3 columns for a Student dataset.

Sample Data:

RollNo	Name	Marks
1	Aarav	78
2	Isha	85
3	Kabir	69
4	Meera	91
5	Zoya	74

Solution:

1. Home tab -> Enter Data.
2. In the grid that opens, rename the default columns to RollNo, Name, Marks.
3. Type the 5 rows of sample values shown.
4. Name the table 'Student' in the Name box, then click Load.

Q7. Explain the difference between Import mode and DirectQuery mode with one example scenario for each.

Solution:

1. Import mode: the entire dataset is copied into Power BI's in-memory engine (VertigoPac). Example: loading a 10,000-row Excel sales file for a monthly report — fast visuals, but data is only as fresh as the last refresh.
2. DirectQuery mode: Power BI sends live queries to the source for every visual instead of storing data. Example: connecting to a large, constantly-updating SQL Server transactional database where near real-time figures are required and the table is too large to import.

Q8. Using a loaded Product Sales dataset, identify the role of the Fields pane, Filters pane, and Visualizations pane.

Sample Data:

ProductID	Category	Sales
P1	Electronics	45000
P2	Furniture	32000
P3	Stationery	8000

Solution:

1. Fields pane: lists ProductID, Category, Sales as draggable items belonging to the loaded table.
2. Visualizations pane: lets you pick a chart type (e.g. Clustered Bar Chart) and, once picked, shows wells for Axis and Values where you drop Category and Sales.
3. Filters pane: lets you add a filter card, e.g. filter the whole page/visual to Sales > 10000, without changing the underlying data.

Q9. Load two data sources — an Excel Employee table and a CSV Department table — into the same Power BI file and confirm both appear in the model.

Sample Data:

DeptID	DeptName
CS	Computer Science
IT	Information Technology
ECE	Electronics

Solution:

1. Get Data -> Excel Workbook -> load Employee table.
2. Get Data -> Text/CSV -> load Department.csv containing DeptID, DeptName.
3. Open the Fields pane and confirm two separate tables, Employee and Department, are both listed.
4. Open Model View to see both tables as separate boxes (no relationship yet).

Q10. Rename the loaded 'Sheet1' table to 'EmployeeData' and rename the column 'Dept' to 'Department' using the interface.

Solution:

1. Go to Data View, right-click 'Sheet1' in the Fields pane -> Rename -> type 'EmployeeData'.
2. In the Data View grid, double-click the 'Dept' column header (or right-click -> Rename) and type 'Department'.
3. Confirm the change is reflected instantly in the Fields pane and in any visuals already using that column.

Set A — Practice Questions

Dataset: Library.csv

BookID	Title	Author	Genre	Price
1	The Alchemist	Paulo Coelho	Fiction	299
2	Sapiens	Yuval Noah Harari	Non-Fiction	450
3	Atomic Habits	James Clear	Self-Help	399
4	Wings of Fire	A.P.J. Abdul Kalam	Biography	250

Questions:

1. Connect to Library.csv using Get Data and load it into Power BI.
2. Identify and label the Ribbon, Fields pane, and Visualizations pane after loading the table.
3. Switch between Report View, Data View, and Model View and note what each displays for the Library table.
4. Decide whether this connection should use Import mode or DirectQuery mode, and justify your choice.
5. Rename the loaded table to 'LibraryCatalog' and rename the 'Price' column to 'PriceINR'.

Set B — Practice Questions

Dataset: Hospital.xlsx (Patients sheet)

PatientID	Name	Age	Department	Doctor
P1	Rakesh Jain	45	Cardiology	Dr. Mehta
P2	Sunita Rao	32	Orthopedics	Dr. Kulkarni
P3	Vikram Singh	58	Cardiology	Dr. Mehta
P4	Anjali Desai	29	Dermatology	Dr. Shah

Questions:

1. Connect to the Hospital.xlsx workbook and load the Patients sheet into Power BI.
2. List the steps needed to load only the Patients sheet if the workbook contains multiple sheets.

3. Compare what the data pane and Filters pane each show once the Patients table is loaded.
4. Use 'Enter Data' to manually create a small 5-row Doctor table and load it alongside Patients.
5. Rename the 'Department' column to 'Speciality' through the interface.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 2: Importing and Cleaning Data from Multiple Data Sources

Objective: Import datasets from Excel, CSV, and a database, then apply basic cleaning operations — removing nulls, renaming fields, and fixing data types — using Power Query Editor.

SET A

Q1. Import an Excel sheet containing employee data with missing Salary values and remove the rows with nulls.

Sample Data:

EmpID	Name	Salary
101	Anita	55000
102	Rahul	null
103	Priya	52000
104	Sameer	null
105	Divya	51000

Solution:

1. Get Data -> Excel Workbook -> load table -> click Transform Data to open Power Query Editor.
2. Click the column header dropdown for 'Salary' -> untick '(blank)' or use Home -> Remove Rows -> Remove Blank Rows.
3. Alternatively right-click the Salary column -> Remove Errors/Remove Empty.
4. Resulting table keeps only EmpID 101, 103, 105.
5. Click Close & Apply.

Q2. Import a CSV Sales file and rename the fields 'Prod' and 'Qty' to more descriptive names.

Sample Data:

Prod	Qty	Amt
Laptop	2	90000
Mouse	5	1500

Solution:

1. Get Data -> Text/CSV -> select sales.csv -> Transform Data.
2. Double-click the 'Prod' header in Power Query -> rename to 'ProductName'.
3. Double-click 'Qty' -> rename to 'Quantity'; rename 'Amt' -> 'Amount'.

4. Close & Apply.

Q3. Connect to a sample SQL Server/Access database and import the 'Customers' table.

Sample Data:

CustomerID	CustomerName	City
C1	Rohan Traders	Pune
C2	Global Mart	Mumbai

Solution:

1. Get Data -> Database -> SQL Server database (or Access database for .accdb).
2. Enter the Server name and Database name, click OK.
3. Choose Import or DirectQuery mode.
4. In Navigator, tick the 'Customers' table and click Load (or Transform Data to clean first).

Q4. Convert the 'OrderDate' column from Text to Date type, and the 'Amount' column from Text to Decimal Number.

Sample Data:

OrderID	OrderDate (text)	Amount (text)
1	01/12/2025	90000
2	03/12/2025	1500.50

Solution:

1. In Power Query Editor, select the OrderDate column.
2. Home tab -> Data Type dropdown -> choose Date. Power Query converts '01/12/2025' to a proper date value.
3. Select the Amount column -> Data Type -> Decimal Number, so '1500.50' becomes a numeric 1500.50 usable in calculations.
4. Close & Apply.

Q5. Combine two Excel tables with the same schema — North Region Sales and South Region Sales — into a single table using Append Queries.

Sample Data:

Region	Product	Sales
North	Laptop	50000
South	Laptop	42000
North	Mouse	1200
South	Mouse	900

Solution:

1. Load 'North_Sales' and 'South_Sales' as two separate queries in Power Query.
2. Home tab -> Append Queries -> Append Queries as New.
3. Select 'North_Sales' as the first table and 'South_Sales' as the table to append.
4. The result is one combined table with a Region column distinguishing the source, as shown in the sample.
5. Rename the new query 'Combined_Sales' and Close & Apply.

SET B

Q6. Detect and remove duplicate rows from a Customer dataset.

Sample Data:

CustomerID	CustomerName
C1	Rohan Traders
C2	Global Mart
C1	Rohan Traders
C3	City Supplies

Solution:

1. In Power Query, select the CustomerID column (or select all columns if the whole row must match).
2. Home tab -> Remove Rows -> Remove Duplicates.
3. The duplicate C1 / Rohan Traders row is removed, leaving 3 unique customers.
4. Close & Apply.

Q7. Handle missing values in an 'Age' column two ways: (a) replace nulls with the column average, and (b) remove rows with nulls.

Sample Data:

StudentID	Age
1	20
2	null
3	22
4	null
5	21

Solution:

1. Approach (a) Replace with average: In Power Query, right-click the Age column header -> Transform -> average may need a manual step: add a custom column with Table.AverageColumn or compute the average via Group By, then use Replace Values to substitute nulls with that average (approx. 21).
2. Approach (b) Remove rows: select Age column -> right-click a null cell -> Remove Empty, or Home -> Remove Rows -> Remove Blank Rows, leaving only StudentID 1, 3, 5.
3. Keep whichever query matches the analysis need; both are valid cleaning strategies depending on context.

Q8. A workbook has 3 sheets — Jan_Sales, Feb_Sales, Mar_Sales — each with the same columns. Import and combine them into one table.

Sample Data:

Month	Product	Sales
Jan	Laptop	40000
Feb	Laptop	45000
Mar	Laptop	50000

Solution:

1. Get Data -> Excel Workbook -> in Navigator tick all three sheets (or use Get Data -> Combine & Transform).
2. In Power Query, select all three queries -> right-click -> Append Queries as New.
3. Add a 'Month' column (using Add Column -> Custom Column) to each before appending if it is not already present, so the combined table can be filtered by month.
4. Close & Apply.

Q9. A 'Marks' column loaded as Text because a few cells contain 'NA'. Fix the data type without losing the NA rows as errors.

Sample Data:

RollNo	Marks
1	78
2	NA
3	91

Solution:

1. Select the Marks column in Power Query -> use Replace Values to change 'NA' to null first.
2. Then Home tab -> Data Type -> Whole Number; because the offending text was already replaced with null, the conversion succeeds instead of producing #Error rows.
3. Optionally use Home -> Detect Data Type after cleaning to confirm Power Query now infers Whole Number automatically.

Q10. Standardize inconsistent text values in a 'Gender' column that contains 'Male', 'M', and 'male'.

Sample Data:

ID	Gender
1	Male
2	M
3	male
4	Female
5	F

Solution:

1. Select the Gender column in Power Query.
2. Use Transform -> Format -> lowercase (or uppercase) to normalise case first, e.g. 'male'.
3. Then Home -> Replace Values -> replace 'm' with 'male' and 'f' with 'female' so all variants collapse to two consistent labels.
4. Close & Apply and verify with a quick Group By count that only 'Male'/'Female' remain.

Set A — Practice Questions

Dataset: Orders (with missing values)

OrderID	CustomerName	City	Amount
1001	Meera Joshi	Pune	5000
1002		Mumbai	3200
1003	Ravi Kumar		4100
1004	Neha Gupta	Nagpur	null

Questions:

1. Import the Orders table above into Power Query and identify every null/blank cell.
2. Remove the row(s) where CustomerName is blank.
3. Replace missing City values with 'Unknown' instead of removing the row.
4. Fix the data type of the Amount column and correct the row where conversion fails.
5. Rename OrderID to 'Order_No' and CustomerName to 'Customer'.

Set B — Practice Questions

Dataset: Inventory (Excel + CSV, with duplicates)

ItemID	ItemName	Category	Stock
I1	Notebook	Stationery	200
I2	Pen	Stationery	500
I1	Notebook	Stationery	200
I3	Marker	stationery	null

Questions:

1. Import ItemMaster.xlsx and ItemPrices.csv (assume both share the ItemID key) and prepare them to be combined.
2. Remove the duplicate Notebook (I1) row.
3. Replace the null Stock value for Marker (I3) with 0.
4. Standardize the Category column so 'stationery' (lowercase) matches 'Stationery'.
5. Explicitly set the ItemID column's data type to Text and confirm the change.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 3: Data Transformation using Power Query

Objective: Apply transformation operations — filtering, merging, appending, splitting columns, creating new columns, grouping, pivoting and unpivoting — using Power Query Editor.

SET A

Q1. Filter a Sales table to keep only rows where Sales is greater than 5000.

Sample Data:

ProductID	Sales
P1	3000
P2	8000
P3	5200
P4	4500

Solution:

1. In Power Query, click the dropdown arrow on the Sales column header.
2. Choose Number Filters -> Greater Than -> enter 5000 -> OK.
3. Result keeps P2 (8000) and P3 (5200); P1 and P4 are removed.

Q2. Merge an Orders table with a Customers table using CustomerID as the key (Left Outer Join) to bring in CustomerName.

Sample Data:

OrderID	CustomerID	CustomerName (after merge)
O1	C1	Rohan Traders
O2	C2	Global Mart
O3	C1	Rohan Traders

Solution:

1. In Power Query, select the Orders query -> Home -> Merge Queries.
2. Pick Customers as the table to merge with, click CustomerID in both tables to match.
3. Set Join Kind = Left Outer (all rows from Orders, matching rows from Customers).
4. Expand the new Customers column and tick only CustomerName.
5. Close & Apply — the Orders table now includes CustomerName as shown.

Q3. Append North and South region sales queries (same columns) into one combined query.

Sample Data:

Region	Product	Sales
North	Chair	12000
South	Chair	9800

Solution:

1. Home tab -> Append Queries -> Append Queries as New.
2. Choose North_Sales as first table, South_Sales as the table to append.
3. The result stacks both tables' rows into one, keeping the Region column to tell them apart.

Q4. Split a 'FullName' column into 'FirstName' and 'LastName' using a space delimiter.

Sample Data:

ID	FullName
1	Anita Sharma
2	Rahul Verma

Solution:

1. Select the FullName column in Power Query.
2. Home tab -> Split Column -> By Delimiter.
3. Choose Space as the delimiter, 'At the leftmost delimiter'.
4. Power Query creates FullName.1 and FullName.2; rename them FirstName and LastName.

Q5. Create a calculated column 'TotalPrice' = Quantity * UnitPrice.

Sample Data:

ProductID	Quantity	UnitPrice
P1	2	500
P2	5	300

Solution:

1. Add Column tab -> Custom Column.
2. Name the new column 'TotalPrice'.

3. Formula: = [Quantity] * [UnitPrice].
4. Click OK; Power Query adds TotalPrice = 1000 for P1 and 1500 for P2.

SET B

Q6. Group an Employee table by Department and calculate the average Salary per department.

Sample Data:

Department	Salary
CS	55000
CS	52000
IT	48000
IT	51000

Solution:

1. Home tab -> Group By.
2. Group by 'Department'.
3. New column name 'AvgSalary', Operation = Average, Column = Salary.
4. Result: CS = 53500, IT = 49500.

Q7. Pivot a table so that values in a 'Month' column (Jan, Feb, Mar) become separate columns holding Sales figures.

Sample Data:

Product	Month	Sales
Laptop	Jan	40000
Laptop	Feb	45000
Laptop	Mar	50000

Solution:

1. Select the 'Month' column in Power Query.
2. Transform tab -> Pivot Column.
3. Values Column = Sales, Aggregate Value Function = Sum (or Don't Aggregate if one row per Product/Month).
4. Result: one row per Product with Jan, Feb, Mar as separate columns holding the corresponding Sales values.

Q8. Unpivot a table that has Jan, Feb, Mar as separate sales columns back into rows (Month, Sales).

Sample Data:

Product	Jan	Feb	Mar
Laptop	40000	45000	50000

Solution:

1. Select the Jan, Feb, Mar columns together (Ctrl+click headers).
2. Transform tab -> Unpivot Columns.
3. Power Query converts them into two columns, 'Attribute' and 'Value'; rename these to 'Month' and 'Sales'.
4. Result: 3 rows for Laptop — Jan/40000, Feb/45000, Mar/50000.

Q9. Extract the Year and Month Name from an OrderDate column.**Sample Data:**

OrderID	OrderDate
1	2025-12-01
2	2026-02-15

Solution:

1. Select the OrderDate column (must already be Date type).
2. Add Column tab -> Date -> Year -> Year, to add an 'OrderDate.Year' column (2025, 2026).
3. Add Column tab -> Date -> Month -> Name of Month, to add an 'OrderDate.MonthName' column ('December', 'February').

Q10. Remove unnecessary columns from a wide dataset and reorder the remaining columns for clarity.**Sample Data:**

OrderID	Customer ID	TempFlag	Notes	OrderDate	Amount
1	C1	x		2025-12-01	90000

Solution:

1. Select the columns to drop (TempFlag, Notes) with Ctrl+click.
2. Right-click -> Remove Columns.
3. Drag column headers (or use Transform -> Move) to reorder remaining columns, e.g. OrderDate, CustomerID, OrderID, Amount.
4. Close & Apply once the layout is confirmed.

Set A — Practice Questions

Dataset: Sales

Region	Product	Qty	UnitPrice
North	Chair	10	1500
South	Table	4	6000
North	Table	2	6000
East	Chair	8	1500

Questions:

1. Filter the dataset to show only rows where Qty is greater than 5.
2. Add a calculated column 'TotalPrice' = Qty * UnitPrice.
3. Group the data by Region and compute the total TotalPrice per region.
4. Split a combined 'Region-Product' text column into two separate columns (assume such a column exists).
5. Pivot the Product column so each distinct product becomes its own column of total Qty sold.

Set B — Practice Questions

Dataset: Employee

EmpID	Name	Dept
E1	Kunal Shah	HR
E2	Priti Rane	Finance

Dataset: Leave

LeaveID	EmpID	LeaveDate	Days
L1	E1	2026-01-10	2
L2	E2	2026-02-05	1

Questions:

1. Merge the Employee and Leave tables on EmpID (Left Outer Join) so each leave record shows the employee's name.
2. Append two months' Leave tables (Jan_Leave and Feb_Leave, same structure) into one combined Leave table.
3. Add a custom column 'LeaveMonth' that extracts the month name from LeaveDate.

4. Unpivot a wide leave-summary table (columns Jan, Feb, Mar holding days) into rows of Month and Days.
5. Remove the Dept column from the merged output and reorder the remaining columns.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 4: Creating Data Models and Establishing Relationships

Objective: Build a data model using multiple tables, define primary-foreign key relationships, construct a star schema, and validate relationships using DAX measures and visuals.

SET A

Q1. Create a data model with Orders and Customers tables and establish a one-to-many relationship on CustomerID.

Sample Data:

Table	Key column
Customers (one)	CustomerID
Orders (many)	CustomerID

Solution:

1. Load both tables, then switch to Model View.
2. Drag CustomerID from Customers onto CustomerID in Orders to create the relationship (or use Modeling -> Manage Relationships -> New).
3. Confirm Cardinality = One to Many, Cross filter direction = Single, with Customers as the 'one' side.

Q2. Given Orders and Products tables, identify the primary key and foreign key.

Sample Data:

Orders	Products
OrderID (PK), ProductID (FK), Qty	ProductID (PK), ProductName, Price

Solution:

1. Primary Key: ProductID in the Products table — uniquely identifies each product.
2. Foreign Key: ProductID in the Orders table — references the Products table to link each order line to a product.
3. OrderID is the primary key of Orders itself.

Q3. Build a star schema with one Sales fact table and three dimension tables: Product, Customer, and Date.

Sample Data:

Sales (fact)	Product (dim)	Customer (dim)	Date (dim)
SalesID, ProductID, CustomerID, DateKey, Amount	ProductID, ProductName, Category	CustomerID, CustomerName, City	DateKey, Date, Month, Year

Solution:

1. Load all four tables into Power BI.
2. In Model View, connect Sales.ProductID -> Product.ProductID, Sales.CustomerID -> Customer.CustomerID, Sales.DateKey -> Date.DateKey, each as One (dimension) to Many (fact).
3. Arrange the fact table in the centre with dimensions around it — the classic star layout.

Q4. Create a relationship between the Sales fact table and the Date dimension table, then test it using a Year slicer.

Sample Data:

Date table	Sales table
DateKey, Year 2024/2025	DateKey, Amount

Solution:

1. In Model View, drag DateKey from Date to DateKey in Sales to form the relationship.
2. In Report View, add a Slicer visual and drop the Year field from the Date table into it.
3. Add a Card or Table visual showing SUM(Sales[Amount]); selecting a year in the slicer should filter the total, confirming the relationship works.

Q5. Two tables, Students and Subjects, have a many-to-many relationship through enrolments. Resolve it using a bridge table.

Sample Data:

Students	Enrolment (bridge)	Subjects
StudentID, Name	StudentID, SubjectID	SubjectID, SubjectName

Solution:

1. Create/import an Enrolment table containing only StudentID and SubjectID (one row per student-subject pair).

2. Relate Students.StudentID -> Enrolment.StudentID (One to Many) and Subjects.SubjectID -> Enrolment.SubjectID (One to Many).
3. This converts the awkward many-to-many into two clean one-to-many relationships through the bridge table.

SET B

Q6. Use the RELATED DAX function to pull ProductName from the Product table into the Sales table as a calculated column.

Sample Data:

Sales	Product
ProductID = P1	ProductID = P1, ProductName = Laptop

Solution:

1. Ensure a relationship exists from Sales.ProductID to Product.ProductID.
2. In the Sales table (Data View), Modeling -> New Column.
3. Formula: ProductName = RELATED(Product[ProductName]).
4. The new column pulls 'Laptop' into every Sales row where ProductID = P1.

Q7. Explain the effect of changing a relationship's cross-filter direction from Single to Both, using a Product-Sales example.

Solution:

1. Single direction (default for one-to-many): filters flow only from Product (one side) to Sales (many side) — filtering by ProductName affects Sales totals, but filtering Sales does not affect the Product list.
2. Both direction: filters flow in both directions — filtering Sales (e.g. Amount > 10000) would now also restrict which Products appear, which is useful in some bidirectional models but can cause ambiguity in complex schemas with multiple paths.
3. Set cross-filter direction via Modeling -> Manage Relationships -> Edit -> Cross filter direction.

Q8. Create a hierarchy Category > Subcategory > Product in the Product table for drill-down analysis.

Sample Data:

Category	Subcategory	Product
Electronics	Computers	Laptop
Electronics	Computers	Mouse
Furniture	Seating	Chair

Solution:

1. In Fields pane, right-click the Category field in the Product table -> Create Hierarchy.
2. Drag Subcategory into the new hierarchy, then drag Product below it, keeping this order.
3. Rename the hierarchy 'Product Hierarchy'; use it on a Matrix or Bar chart visual to drill from Category down to Product.

Q9. Using Manage Relationships, correct a relationship that was wrongly set as Many-to-Many when it should be One-to-Many.

Solution:

1. Modeling tab -> Manage Relationships.
2. Select the incorrect relationship and click Edit.
3. Change Cardinality from Many-to-Many to One-to-Many, ensuring the 'one' side truly has unique key values (e.g. Customer.CustomerID).
4. Click OK and re-validate visuals that use this relationship for correct totals.

Q10. Create a DAX measure 'Total Sales = SUM(Sales[Amount])' and verify the model by showing it against Customer Name in a table visual.

Sample Data:

CustomerName	Total Sales
Rohan Traders	132000
Global Mart	98000

Solution:

1. In the Sales table, Modeling -> New Measure.
2. Formula: Total Sales = SUM(Sales[Amount]).
3. Add a Table visual, drag CustomerName from Customer table and Total Sales measure into it.
4. Correct per-customer totals confirm the Sales-to-Customer relationship is working properly.

Set A — Practice Questions

Dataset: Products

ProductID	ProductName	Category
P1	Laptop	Electronics
P2	Chair	Furniture

Dataset: Customers

CustomerID	CustomerName	City
C1	Rohan Traders	Pune
C2	Global Mart	Mumbai

Dataset: Orders

OrderID	ProductID	CustomerID	Qty
O1	P1	C1	2
O2	P2	C2	5

Questions:

1. Load Products, Customers, and Orders, and identify the primary key of each table.
2. Create One-to-Many relationships: Products-Orders and Customers-Orders.
3. Describe the star schema formed here, naming the fact table and the dimension tables.
4. Use RELATED() to pull ProductName and CustomerName into the Orders table as calculated columns.
5. Create a measure 'Total Qty Sold = SUM(Orders[Qty])' and validate it in a table visual against ProductName.

Set B — Practice Questions

Dataset: Students

StudentID	StudentName
S1	Aarav
S2	Isha

Dataset: Courses

CourseID	CourseName
CR1	DBMS
CR2	Data Analytics

Dataset: Enrolment

StudentID	CourseID
S1	CR1
S1	CR2
S2	CR1

Questions:

1. Explain why a direct relationship between Students and Courses would be many-to-many.
2. Create the Enrolment bridge-table relationships: Students-Enrolment and Courses-Enrolment, both One-to-Many.
3. Build a simple hierarchy using CourseID and CourseName for drill-down.
4. Set the cross-filter direction on one relationship to 'Both' and explain the expected impact on visuals.
5. Create a measure 'Total Enrolments = COUNTROWS(Enrolment)' and display it by CourseName in a table visual.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 5: Designing Basic Charts for Data Visualization

Objective: Create and customize at least four chart types (bar, line, pie, area, and others) using a sample dataset, applying titles, axis labels, and formatting.

SET A

Q1. Create a Clustered Bar Chart of Sales by Region.

Sample Data:

Region	Sales
North	120000
South	95000
East	80000
West	110000

Solution:

1. Select the Bar chart visual from the Visualizations pane.
2. Drag Region to Axis and Sales to Values.
3. Format pane -> Title -> type 'Sales by Region'.
4. Format pane -> X-axis/Y-axis -> adjust title text to 'Region' and 'Total Sales'.

Q2. Create a Line Chart showing the Monthly Sales Trend for a year.

Sample Data:

Month	Sales
Jan	40000
Feb	45000
Mar	50000
Apr	47000

Solution:

1. Select the Line chart visual.
2. Drag Month to the X-axis and Sales to Values.
3. Ensure Month is sorted chronologically (right-click column -> Sort by Column if a MonthNumber field exists).
4. Format -> Title -> 'Monthly Sales Trend'.

Q3. Create a Pie Chart showing Market Share by Product Category.

Sample Data:

Category	Sales
Electronics	150000
Furniture	70000
Stationery	30000

Solution:

1. Select the Pie chart visual.
2. Drag Category to Legend and Sales to Values.
3. Format -> Data labels -> turn On, set to show Percentage of total.
4. Format -> Title -> 'Market Share by Category'.

Q4. Create an Area Chart comparing Revenue over time for two years (2024 vs 2025).

Sample Data:

Month	2024	2025
Jan	38000	40000
Feb	41000	45000
Mar	44000	50000

Solution:

1. Reshape data so Year is a column value (unpivot 2024/2025 into Year, Revenue) if not already in that form.
2. Select the Area chart visual (or Stacked Area chart).
3. Drag Month to X-axis, Revenue to Values, and Year to Legend to compare the two years as separate shaded series.
4. Format -> Title -> 'Revenue Trend: 2024 vs 2025'.

Q5. Customize the title, axis labels, and data labels of the Bar Chart created in Q1.

Solution:

1. Select the bar chart -> open the Format pane (paint-roller icon).
2. Under 'General' or 'Title', set Title text to 'Regional Sales Performance', font size 14, bold.
3. Under 'X-axis' and 'Y-axis', enable and edit the Axis title text ('Region', 'Sales (₹)').
4. Under 'Data labels', turn On, choose position 'Outside end', and set display units (e.g. Thousands).

SET B

Q6. Create a Stacked Bar Chart showing Sales by Region broken down by Product Category.

Sample Data:

Region	Category	Sales
North	Electronics	70000
North	Furniture	50000
South	Electronics	55000
South	Furniture	40000

Solution:

1. Select the Stacked Bar Chart visual.
2. Drag Region to Axis, Sales to Values, and Category to Legend.
3. Each bar (Region) now shows segments stacked by Category, e.g. North splits into Electronics 70000 and Furniture 50000.
4. Format -> Title -> 'Sales by Region and Category'.

Q7. Apply conditional formatting to a bar chart so that bars above a target Sales value appear in a different color.

Sample Data:

Region	Sales
North	120000
South	95000
East	80000
West	110000

Solution:

1. Select the bar chart -> Format pane -> Columns -> Colors.
2. Click the fx (conditional formatting) icon next to Color.
3. Set rule: If Sales >= 100000, color = Green; else color = Grey.
4. Click OK; bars for North and West turn green, South and East stay grey.

Q8. Add a trend line to a Scatter chart plotting Advertising Spend vs Sales.

Sample Data:

AdSpend	Sales
2000	40000
3000	52000
4000	61000
5000	75000

Solution:

1. Select the Scatter chart visual; drag AdSpend to X-axis, Sales to Y-axis.
2. With the visual selected, go to the Analytics pane (magnifying-glass icon).
3. Under 'Trend line', click Add — Power BI fits and draws a linear trend line through the points.
4. Format the trend line's color/style if needed under the same Analytics pane.

Q9. Create a Combo Chart (bar + line) showing Sales as bars and Profit Margin % as a line on a secondary axis.

Sample Data:

Month	Sales	ProfitMargin%
Jan	40000	18
Feb	45000	20
Mar	50000	22

Solution:

1. Select the 'Line and Stacked Column Chart' (or 'Line and Clustered Column Chart') visual.
2. Drag Month to Shared Axis, Sales to Column Values, ProfitMargin% to Line Values.
3. Power BI automatically places ProfitMargin% on a secondary (right) Y-axis since its scale differs from Sales.
4. Format -> Title -> 'Sales vs Profit Margin'.

Q10. Format the legend position and font size on the Pie Chart from Q3 for better readability.

Solution:

1. Select the pie chart -> Format pane -> Legend.
2. Set Position to 'Right' (or 'Bottom' if the chart is wide and short).
3. Increase Text Size to 12-14pt and set Font Family for consistency with the rest of the report.

- Toggle Title within the legend section if a legend header is wanted.

Set A — Practice Questions

Dataset: Quarterly Sales

Quarter	Region	Sales
Q1	North	120000
Q1	South	95000
Q2	North	130000
Q2	South	100000

Questions:

- Create a Clustered Bar chart of Sales by Region.
- Create a Line chart showing the Sales trend across Quarters.
- Create a Pie chart of Sales share by Region for Q1 only.
- Customize the bar chart's title, axis labels, and data labels.
- Create a Stacked Column chart of Sales by Quarter and Region.

Set B — Practice Questions

Dataset: Website Traffic

Month	Visitors	Conversions
Jan	10000	400
Feb	12000	550
Mar	15000	700

Questions:

- Create an Area chart of Visitors over the three months.
- Create a Combo chart (bar for Visitors, line for Conversions) on a shared Month axis.
- Create a Scatter chart of Visitors vs Conversions and add a trend line.
- Format the legend position and font size for the Combo chart.
- Apply conditional formatting so bars are colored differently where Conversions exceed 500.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 6: Using Filters, Slicers, and Parameters to Enhance Interactivity

Objective: Add interactive components — visual/page/report-level filters, slicers, What-if parameters and field parameters — to enable dynamic, user-driven selection of data categories.

SET A

Q1. Apply a visual-level filter so a Bar Chart shows only products where Sales is greater than 5000.

Sample Data:

Product	Sales
Laptop	9000
Mouse	1500
Chair	6200
Pen	300

Solution:

1. Select the bar chart visual showing Product vs Sales.
2. Open the Filters pane -> drag Sales into the 'Filters on this visual' well.
3. Set filter type to 'Basic' or 'Advanced' -> condition 'is greater than' 5000 -> Apply filter.
4. Only Laptop (9000) and Chair (6200) remain visible in the chart.

Q2. Apply a page-level filter that restricts an entire report page to the 'North' region only.

Sample Data:

Region	Sales
North	50000
South	42000
East	30000

Solution:

1. Open the Filters pane and drag the Region field into 'Filters on this page'.
2. Set filter type to Basic filtering, tick only 'North'.
3. Every visual on that page now reflects North-region data only, regardless of individual visual filters.

Q3. Apply a report-level filter to restrict the whole report to Year = 2025.

Sample Data:

Year	Sales
2024	400000
2025	520000

Solution:

1. Open the Filters pane -> drag the Year field into 'Filters on all pages'.
2. Set Basic filtering, tick only '2025'.
3. All pages across the report now show 2025 data only.

Q4. Add a Slicer for 'Category' to a report page and observe how selecting a value cross-filters other visuals.

Sample Data:

Category	Product	Sales
Electronics	Laptop	9000
Electronics	Mouse	1500
Furniture	Chair	6200

Solution:

1. Insert a Slicer visual and drag Category into its Field well.
2. Place a Bar chart of Product vs Sales on the same page.
3. Click 'Electronics' in the slicer; the bar chart automatically filters to show only Laptop and Mouse, demonstrating cross-filtering.

Q5. Add a Slicer that lets the user pick a date range (between two dates) using an OrderDate field.

Sample Data:

OrderID	OrderDate
1	2026-01-05
2	2026-02-10
3	2026-03-20

Solution:

1. Insert a Slicer visual, drag OrderDate into the Field well.
2. In the Slicer's Format pane, set 'Slicer settings -> Options -> Style' to 'Between'.

3. The slicer now shows two date pickers/a range slider; selecting 2026-01-01 to 2026-02-28 filters out OrderID 3.

SET B

Q6. Create a 'What-if' Parameter for Discount % (0% to 30%, step 5%) and use it to dynamically adjust Total Price.

Sample Data:

Product	Price
Laptop	90000
Chair	6000

Solution:

1. Modeling tab -> New Parameter -> What-If Parameter.
2. Name: 'Discount %'; Data type: Decimal; Min 0, Max 0.30, Increment 0.05; this auto-creates a Discount % table, a slicer, and a Discount % (value) measure.
3. Create a measure: DiscountedPrice = SUM(Products[Price]) * (1 - 'Discount %'[Discount % Value]).
4. Add the auto-generated slicer to the page; moving it recalculates DiscountedPrice live, e.g. 90000 becomes 76500 at 15% discount.

Q7. Create a Field Parameter that lets the user switch a chart between showing Sales and Profit.

Sample Data:

Region	Sales	Profit
North	120000	24000
South	95000	15000

Solution:

1. Modeling tab -> New Parameter -> Fields.
2. Name it 'Metric Selector'; add the Sales and Profit fields to the parameter list; tick 'Add slicer to this page'.
3. Drag the resulting 'Metric Selector' field into the Values well of a bar chart instead of Sales/Profit directly.
4. The accompanying slicer now toggles the chart between Sales and Profit views.

Q8. Sync a Region slicer across two report pages so the selection stays consistent when the user switches pages.

Solution:

1. Add the Region slicer to Page 1.
2. View tab -> Sync Slicers to open the Sync slicers pane.
3. On Page 2, tick both 'Sync' and 'Visible' checkboxes for the same slicer.
4. Selecting 'South' on Page 1 automatically applies the same filter when the user navigates to Page 2.

Q9. Apply a Top N filter to display only the Top 5 products by Sales in a table visual.

Sample Data:

Product	Sales
Laptop	90000
Chair	42000
Mouse	15000
Pen	3000
Table	55000
Sofa	61000

Solution:

1. Select the table/bar visual with Product and Sales.
2. Drag Product into the Filters pane, set filter type to 'Top N'.
3. Show items: Top 5, By value: Sales -> Apply filter.
4. Only the 5 highest-selling products remain, Pen (lowest at 3000) is dropped.

Q10. Set up a Drillthrough page that shows detailed Order records when a user right-clicks a Region on the summary page.

Sample Data:

OrderID	Region	Amount
O1	North	5000
O2	North	3200
O3	South	4100

Solution:

1. Create a new page named 'Region Details' and add a table visual of OrderID, Region, Amount.
2. Drag Region into the 'Drillthrough' well in the Filters pane on that page; this adds a back button automatically.
3. On the summary page, right-click a Region value in any visual -> Drillthrough -> Region Details.

4. The details page opens filtered to only that Region's orders.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 7: Creating Calculated Fields, Measures, and KPIs

Objective: Create calculated columns, DAX measures (SUM, AVERAGE, COUNT, CALCULATE), time-intelligence measures, and KPI visuals to surface key business metrics.

SET A

Q1. Create a calculated column 'Profit' = Revenue - Cost.

Sample Data:

OrderID	Revenue	Cost
O1	10000	7000
O2	5000	4200

Solution:

1. In Data View, select the Sales table -> Modeling -> New Column.
2. Formula: Profit = [Revenue] - [Cost].
3. Result: O1 -> 3000, O2 -> 800, added as a new column.

Q2. Create a measure 'Total Sales' using SUM.

Sample Data:

OrderID	Amount
O1	10000
O2	5000
O3	7500

Solution:

1. Modeling tab -> New Measure.
2. Formula: Total Sales = SUM(Sales[Amount]).
3. Result when placed in a Card visual: 22500.

Q3. Create a measure 'Average Order Value' using AVERAGE.

Sample Data:

OrderID	Amount
O1	10000
O2	5000
O3	7500

Solution:

1. Modeling tab -> New Measure.
2. Formula: Average Order Value = AVERAGE(Sales[Amount]).
3. Result: $(10000+5000+7500)/3 = 7500$.

Q4. Create a measure 'Order Count' using COUNT / COUNTROWS.

Sample Data:

OrderID	Amount
O1	10000
O2	5000
O3	7500

Solution:

1. Modeling tab -> New Measure.
2. Formula: Order Count = COUNTROWS(Sales) (or COUNT(Sales[OrderID])).
3. Result: 3, i.e. the number of order rows in the table.

Q5. Create a measure that returns total Sales only for the 'North' region using CALCULATE and FILTER.

Sample Data:

Region	Amount
North	10000
South	5000
North	3000

Solution:

1. Modeling tab -> New Measure.
2. Formula: North Sales = CALCULATE(SUM(Sales[Amount]), Sales[Region] = "North").
3. Result: $10000 + 3000 = 13000$, filtering out South rows.

SET B

Q6. Use the Quick Measure feature to create a Running Total of Sales over Date.

Sample Data:

Date	Amount
2026-01-01	1000
2026-01-02	1500
2026-01-03	1200

Solution:

1. Right-click the Sales table (or use Calculations group) -> New Quick Measure.
2. Choose calculation type 'Running Total', Base value = Amount, set the Date field as the running-total axis.
3. Power BI generates the DAX automatically; result shows a cumulative total: 1000, 2500, 3700.

Q7. Create a KPI visual comparing Actual Sales against a Target Sales value.

Sample Data:

Month	ActualSales	TargetSales
Jan	45000	50000
Feb	52000	50000

Solution:

1. Create two measures: Actual Sales = SUM(Sales[ActualSales]) and Target Sales = SUM(Sales[TargetSales]).
2. Insert the KPI visual; set Indicator = Actual Sales, Trend axis = Month, Target goals = Target Sales.
3. The KPI shows Feb trending above target (green) and Jan below target (red/amber), depending on the configured direction.

Q8. Create a calculated column that classifies each product as 'High' or 'Low' performer based on a Sales threshold of 50000.

Sample Data:

Product	Sales
Laptop	90000
Pen	3000

Solution:

1. Modeling -> New Column on the Sales/Product table.
2. Formula: Performance = IF([Sales] >= 50000, "High", "Low").
3. Result: Laptop -> High, Pen -> Low.

Q9. Create a Year-over-Year growth measure using DAX time-intelligence (SAMEPERIODLASTYEAR).

Sample Data:

Date	Amount
2025-03-01	40000
2026-03-01	52000

Solution:

1. Ensure a proper Date table exists and is marked as a Date table, related to Sales.
2. Create: Sales PY = CALCULATE(SUM(Sales[Amount]), SAMEPERIODLASTYEAR('Date'[Date])).
3. Create: YoY Growth % = DIVIDE([Total Sales] - [Sales PY], [Sales PY]).
4. For March 2026 vs March 2025: $(52000 - 40000) / 40000 = 30\%$ growth.

Q10. Create a measure showing each Region's Sales as a percentage of total Sales, using DIVIDE and ALL.

Sample Data:

Region	Amount
North	30000
South	20000
East	10000

Solution:

1. Create: Total Sales = SUM(Sales[Amount]).
2. Create: Grand Total Sales = CALCULATE([Total Sales], ALL(Sales[Region])).
3. Create: % of Total = DIVIDE([Total Sales], [Grand Total Sales]).
4. In a table by Region: North = $30000 / 60000 = 50\%$, South = 33.3% , East = 16.7% .

Set A — Practice Questions

Dataset: Orders

OrderID	Amount	Region
01	5000	North
02	3200	South
03	7000	North

Questions:

1. Create a measure 'Total Orders' using COUNTROWS.
2. Create a measure 'Average Order Value' using AVERAGE.
3. Create a measure that returns total Amount only for the 'North' region using CALCULATE.
4. Create a calculated column classifying each order as 'Large' (Amount >= 5000) or 'Small'.
5. Create a KPI visual comparing Actual Order Count against a Target Order Count.

Set B — Practice Questions***Dataset: Employee***

EmpID	Salary	Dept	JoinDate
E1	55000	CS	2022-06-01
E2	48000	IT	2023-01-15
E3	52000	CS	2021-09-10

Questions:

1. Create a measure 'Total Salary' using SUM.
2. Create a measure 'Headcount' using COUNTROWS(Employee).
3. Create a measure for 'Average Salary' filtered to the CS department using CALCULATE.
4. Create a calculated column 'Tenure (Years)' using DATEDIFF or TODAY() - JoinDate.
5. Create a Year-over-Year headcount growth measure using time-intelligence functions.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 8: Designing Dashboards for Business Insights

Objective: Arrange multiple visuals into a cohesive dashboard, apply themes and formatting, and configure interactions and navigation for an effective business-insight layout.

SET A

Q1. Arrange a Bar chart, Line chart, Pie chart, and Card visual on a single dashboard page for a Sales overview.

Sample Data:

Region	Month	Sales
North	Jan	40000
South	Jan	32000
North	Feb	45000

Solution:

1. Add a Card visual for Total Sales (measure) at the top-left as a headline KPI.
2. Add a Bar chart (Region vs Sales) and a Line chart (Month vs Sales) side by side below the card.
3. Add a Pie chart (Region share of Sales) in a remaining corner.
4. Resize and align all four visuals using the alignment gridlines so edges line up neatly.

Q2. Apply a built-in theme to give the entire dashboard a consistent color scheme.

Solution:

1. View tab -> Themes gallery.
2. Select a built-in theme (e.g. 'Executive' or 'City Park') or a custom JSON theme.
3. All visuals on every page instantly re-color to match the theme's palette, fonts, and backgrounds.

Q3. Customize the dashboard's page background color and add subtle borders around each visual.

Solution:

1. Right-click empty canvas -> Format page -> Canvas background -> choose a light grey or brand color, set transparency.
2. Select each visual -> Format pane -> Effects -> Border -> turn On, choose a thin border color/width for a card-like appearance.

Q4. Add a Card visual displaying Total Revenue as a headline KPI number, and format its font size and color.

Sample Data:

OrderID	Revenue
O1	50000
O2	32000

Solution:

1. Insert a Card visual, drag the Total Revenue measure into the Fields well.
2. Format pane -> Callout value -> increase font size (e.g. 40pt) and set a brand color.
3. Format pane -> Category label -> customize the label text/size beneath the number.

Q5. Create a Bookmark that toggles between a 'Summary View' and a 'Detailed View' of the same dashboard page using a button.

Solution:

1. Arrange the Summary visuals; View tab -> Bookmarks pane -> Add, name it 'Summary View'.
2. Show the Detailed visuals instead (hide/show visuals as needed); Add another bookmark 'Detailed View'.
3. Insert -> Buttons -> Blank button; under Action, set Type = Bookmark and select the relevant bookmark for each button.
4. Clicking each button now switches the dashboard between the two saved states.

SET B

Q6. Add a text box with a dashboard title and a company logo image to the report page.

Solution:

1. Insert tab -> Text box -> type 'Sales Performance Dashboard' -> format font size/color.
2. Insert tab -> Image -> browse and select the logo file -> position it in a top corner and resize proportionally.

Q7. Turn off the cross-filter interaction between a Slicer and one specific chart on the dashboard.

Solution:

1. Select the Slicer visual.

2. Format tab (ribbon) -> Edit Interactions.
3. On the target chart that should NOT be affected, click the 'None' icon that appears above it.
4. That chart will now stay unfiltered when the slicer selection changes, while other visuals still respond.

Q8. Align and evenly distribute four visuals on the canvas using Power BI's alignment tools.

Solution:

1. Select all four visuals (Ctrl+click or drag-select).
2. Format tab -> Align -> choose 'Align top' (or middle) to line up their tops.
3. Format tab -> Distribute -> 'Distribute horizontally' to space them evenly across the page width.

Q9. Customize a report page tooltip so hovering over a bar in a chart shows extra context (e.g. Profit Margin).

Sample Data:

Region	Sales	ProfitMargin%
North	120000	20
South	95000	18

Solution:

1. Create a new page, set its Page information -> 'Allow use as tooltip' = On, and 'Tooltip size' as needed.
2. Add a Card/Table visual on this tooltip page showing ProfitMargin%.
3. On the main bar chart, Format pane -> Tooltip -> Type = Report page -> select this new page.
4. Hovering over a bar now shows the custom tooltip page content instead of the default box.

Q10. Configure a Mobile layout for the dashboard so it displays correctly on a phone screen.

Solution:

1. View tab -> Mobile layout.
2. Drag the visuals from the visual list on the left into the vertical phone-shaped canvas in the order they should appear.
3. Resize each visual within the mobile canvas as needed; this layout is used automatically when the report is opened in the Power BI mobile app.

Set A — Practice Questions

Dataset: Retail Dashboard Visuals

Visual	Metric
Card	Total Sales
Bar Chart	Sales by Region
Line Chart	Sales Trend
Pie Chart	Category Share

Questions:

1. Arrange the four visuals listed (Card, Bar, Line, Pie) into a single retail dashboard page.
2. Apply a built-in theme so all visuals share a consistent color scheme.
3. Add a Bookmark that toggles between a 'Summary View' and a 'Detailed View' using buttons.
4. Format the Card visual's font size and color to stand out as the headline KPI.
5. Turn off the cross-filter interaction between a Slicer and one specific chart on the page.

Set B — Practice Questions

Dataset: HR Dashboard Visuals

Visual	Metric
Card	Total Headcount
Bar Chart	Headcount by Department
Line Chart	Attrition Trend
Table	Average Tenure

Questions:

1. Arrange the four visuals listed into a single HR dashboard page.
2. Customize the dashboard's background color and add subtle borders around each visual.
3. Add a company logo image and a title text box to the dashboard.
4. Align and evenly distribute all four visuals using Power BI's alignment tools.
5. Configure a Mobile layout so the HR dashboard displays correctly on a phone.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 9: Publishing and Sharing Dashboards on Power BI Service

Objective: Publish a completed report to Power BI Service, create and share a dashboard, configure refresh, security, and explore embed/export options for distributing insights.

SET A

Q1. Publish a Power BI Desktop report (.pbix) to a workspace on Power BI Service.

Solution:

1. In Power BI Desktop, Home tab -> Publish.
2. Sign in if prompted, then choose the destination workspace (e.g. 'Sales Analytics').
3. Wait for the upload to complete and click the link shown to open the report in the Service.

Q2. Create a Dashboard in Power BI Service by pinning visuals from the published report.

Sample Data:

Visual	Source page
Total Sales Card	Overview
Sales by Region Bar	Overview

Solution:

1. Open the published report in Power BI Service.
2. Hover over a visual (e.g. Total Sales Card) -> click the pin icon.
3. Choose 'New dashboard', name it 'Executive Dashboard', and pin it.
4. Repeat for the Sales by Region bar chart, pinning to the same dashboard so both visuals sit together.

Q3. Share the newly created dashboard with a colleague, granting them view-only access.

Solution:

1. Open the dashboard -> click Share (top-right).
2. Enter the colleague's email address.
3. Leave 'Allow recipients to share' unchecked if view-only is required, and ensure their access level is Viewer, not Member/Contributor.
4. Click Grant access; they receive an email with a direct link.

Q4. Configure a scheduled refresh so the published dataset updates automatically every day.

Solution:

1. In the workspace, go to the dataset's Settings.
2. Expand 'Scheduled refresh' -> turn it On.
3. Set refresh frequency (e.g. Daily) and time(s), and enter data source credentials if prompted.
4. Save; the dataset will now refresh automatically without reopening Power BI Desktop.

Q5. Publish a report to the web using 'Publish to web' and note the privacy implications.

Solution:

1. In Power BI Service, open the report -> File -> Embed report -> Publish to web (public).
2. Confirm the warning dialog acknowledging the report becomes publicly accessible to anyone with the link.
3. Copy the generated iframe/link code for embedding in a public website.
4. Note: this should never be used for confidential or internal business data since no login is required to view it.

SET B

Q6. Create an App from a workspace and publish it for wider distribution inside the organization.

Solution:

1. In the workspace, click 'Create app'.
2. Fill in the app's Details (name, description, logo) and Navigation (choose which reports/dashboards to include).
3. Under Permissions, choose which users or groups in the organization can access the app.
4. Click Publish app; users can then install it from the Power BI Apps section.

Q7. Configure Row-Level Security (RLS) roles before publishing so each Regional Manager only sees their own region's data.

Sample Data:

Role	DAX filter
North_Manager	[Region] = "North"
South_Manager	[Region] = "South"

Solution:

1. In Power BI Desktop, Modeling tab -> Manage Roles.
2. Create role 'North_Manager' with table filter: [Region] = "North"; create 'South_Manager' similarly.
3. Publish the report, then in Power BI Service, dataset Settings -> Security -> assign each manager's account/email to the matching role.

Q8. Export a published report page to PDF and to PowerPoint directly from Power BI Service.

Solution:

1. Open the report in Power BI Service -> File -> Export to -> PDF; choose to export all pages or current page, then download.
2. Repeat via File -> Export to -> PowerPoint to get an editable slide version with one visual snapshot per slide.

Q9. Embed a published Power BI report inside a Microsoft Teams channel or SharePoint page.

Solution:

1. In Power BI Service, open the report -> File -> Embed report -> SharePoint Online (or use the Power BI app inside Teams).
2. Copy the generated embed link.
3. In SharePoint, add a 'Power BI' web part to the page and paste the report link; in Teams, add a Power BI tab and select the report.

Q10. Manage workspace access so one user gets Contributor rights (can edit) while another gets Viewer rights (read-only).

Solution:

1. Open the workspace -> Access (or workspace Settings -> Access).
2. Add the first user's account and set their role to 'Contributor'.
3. Add the second user's account and set their role to 'Viewer'.
4. Save; the Contributor can edit/publish content, while the Viewer can only open and interact with existing reports/dashboards.

Set A — Practice Questions

Dataset: Sales_Report.pbix (conceptual)

Item	Detail
Workspace	Sales Analytics
Dataset	Sales_Report
Owner	You

Questions:

1. Publish the Sales_Report to the 'Sales Analytics' workspace on Power BI Service.
2. Pin key visuals from the published report to create a new Dashboard.
3. Share the dashboard with a colleague, granting Viewer (view-only) access.
4. Configure a daily scheduled refresh for the published dataset.
5. Export the report to PDF directly from Power BI Service.

Set B — Practice Questions

Dataset: HR_Report.pbix (conceptual)

Item	Detail
Workspace	HR Analytics
Dataset	HR_Report
Roles	Dept_Manager

Questions:

1. Configure Row-Level Security roles so each department manager sees only their own department's data.
2. Create an App from the HR Analytics workspace and publish it to the organization.
3. Embed the HR_Report inside a Microsoft Teams channel.
4. Manage workspace access so one user is a Contributor and another is a Viewer.
5. Use 'Publish to web' on a non-sensitive sample page and explain the privacy risk this introduces.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 10: Integrating Power BI with Excel

Objective: Use Excel as both a data source and an analysis companion for Power BI — via Power Pivot, Analyze in Excel, exporting visual data, and publishing Excel data models to the Service.

SET A

Q1. Import an Excel file that already contains a PivotTable/PivotChart directly into Power BI.

Sample Data:

Region	Sales
North	120000
South	95000

Solution:

1. Get Data -> Excel Workbook -> select the file.
2. In Navigator, tick the underlying data sheet/table (not the PivotTable itself, since Power BI reads the source data range).
3. Click Load; rebuild any needed visuals natively in Power BI using this data.

Q2. Use the 'Analyze in Excel' feature to explore a published Power BI dataset from an Excel PivotTable.

Solution:

1. In Power BI Service, open the semantic model/dataset -> click 'Analyze in Excel' (or from the ellipsis menu).
2. Download and open the generated .xlsx file, which contains a live PivotTable connection (OLEDB) to the Power BI dataset.
3. Drag fields into the PivotTable in Excel; it queries the live Power BI model rather than static data.

Q3. Load two related tables into Excel's Power Pivot data model and create a relationship between them.

Sample Data:

Orders	Customers
OrderID, CustomerID, Amount	CustomerID, CustomerName

Solution:

1. In Excel, Data tab -> Get Data (or Power Pivot tab -> Manage) -> load both tables into the Data Model.
2. In the Power Pivot window, Diagram View -> drag CustomerID from Orders to CustomerID in Customers to create the relationship.
3. Build a PivotTable using fields from both tables to confirm the relationship works.

Q4. Export the underlying data of a Power BI bar chart visual back to an Excel (.xlsx) file.

Sample Data:

Solution:

1. Hover over the visual -> click the More options (...) menu -> Export data.
2. Choose 'Data with current layout' or 'Summarized data', and format = .xlsx.
3. Open the downloaded file in Excel to further analyze or share the numbers.

Q5. Create a DAX measure inside Excel's Power Pivot model, similar to a Power BI measure.

Sample Data:

OrderID	Amount
O1	5000
O2	7000

Solution:

1. In Excel, Power Pivot tab -> Manage -> select the Orders table.
2. In the calculation area at the bottom of the Power Pivot window, type: Total Sales := SUM(Orders[Amount]).
3. Use this measure in a PivotTable's Values area just like a Power BI DAX measure.

SET B

Q6. Use Excel's Get & Transform (Power Query in Excel) to clean an imported CSV, mirroring the Power BI Query Editor workflow.

Sample Data:

ID	Name	Age
1	Rahul	28
2		null

Solution:

1. In Excel, Data tab -> Get Data -> From File -> From Text/CSV -> Transform Data to open Power Query in Excel.
2. Remove the blank Name row and fix the Age column's data type, exactly as done in Power BI's Power Query Editor.
3. Home -> Close & Load to bring the cleaned table into an Excel worksheet or the Data Model.

Q7. Publish an Excel workbook that contains a Power Pivot data model to Power BI Service.**Solution:**

1. Save the Excel workbook (with its Power Pivot model) to OneDrive or SharePoint.
2. In Power BI Service, click Get Data -> Files -> OneDrive/SharePoint -> select the workbook.
3. Power BI Service imports the Excel data model as a dataset, ready to build reports from directly in the browser.

Q8. Set up an Excel file stored in OneDrive as a Power BI data source with scheduled refresh, so edits in Excel flow into the Power BI report.**Solution:**

1. Store the Excel file in OneDrive for Business or SharePoint (not a local drive).
2. In Power BI Desktop, Get Data -> Excel -> connect using the OneDrive/SharePoint URL rather than a local file path.
3. Publish the report, then in the dataset settings enable scheduled refresh; because the source is cloud-based, refresh can run without a local gateway.

Q9. Compare and reconcile the Total Sales figure shown in an Excel PivotTable against the same figure in a Power BI report, for validation.**Sample Data:**

Source	TotalSales
Excel PivotTable	227500
Power BI Card	227500

Solution:

1. Build the same aggregation (Sum of Amount) in both an Excel PivotTable and a Power BI Card visual, from the same underlying table.

2. Compare the two totals; if they differ, check for filters applied only in one tool, or duplicate/missing rows introduced during import.
3. In this example both show 227500, confirming the datasets and calculations match.

Q10. Copy a DAX measure formula created in Power BI into Excel's Power Pivot model and confirm it produces the same result.

Sample Data:

OrderID	Amount
O1	5000
O2	7000
O3	3000

Solution:

1. In Power BI, note the exact formula: Total Sales = SUM(Sales[Amount]).
2. In Excel Power Pivot's calculation area, create an identical measure: Total Sales := SUM(Sales[Amount]).
3. Add each to a table/PivotTable; both should return 15000, confirming DAX behaves consistently across the two tools.

Set A — Practice Questions

Dataset: Sales.xlsx (with Power Pivot model)

OrderID	Amount
O1	5000
O2	7200

Questions:

1. Import the underlying data behind an existing Excel PivotTable into Power BI.
2. Use 'Analyze in Excel' to explore a published Power BI dataset from an Excel PivotTable.
3. Export the data behind a Power BI visual back into an Excel (.xlsx) file.
4. Create a Power Pivot DAX measure in Excel equivalent to a Power BI measure (e.g. Total Sales).
5. Reconcile the Total Sales figure between the Excel PivotTable and the Power BI report.

Set B — Practice Questions

Dataset: Inventory.xlsx

ItemID	ItemName	Stock
I1	Notebook	200
I2	Pen	500

Questions:

1. Load two related Excel tables (Inventory and Suppliers) into Excel's Power Pivot model and create a relationship.
2. Use Power Query in Excel to clean an Inventory CSV (remove nulls, fix data types).
3. Publish an Excel workbook containing a Power Pivot data model to Power BI Service.
4. Set up an Excel file stored in OneDrive as a Power BI source with scheduled refresh.
5. Copy a DAX formula created in Power BI into Excel's Power Pivot model and confirm the result matches.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date:

Practical 11: Mini Project — Interactive Dashboard on a Real-World Dataset

Objective: Apply the full Power BI workflow — data import, cleaning, modeling, DAX, visualization, interactivity, and publishing — end-to-end on a real-world dataset to produce a polished, insight-driven dashboard.

SET A

Q1. Select and import a real-world dataset (e.g. a retail 'Superstore' sales dataset) into Power BI.

Sample Data:

OrderID	OrderDate	Category	Region	Sales	Profit
O1	2025-01-05	Furniture	North	4500	600
O2	2025-01-12	Electronics	South	9200	1500
O3	2025-02-03	Furniture	East	3100	-200

Solution:

1. Get Data -> Excel/CSV -> select the Superstore-style dataset -> Transform Data.
2. Confirm all columns (OrderID, OrderDate, Category, Region, Sales, Profit) load with sensible data types.
3. Close & Apply to bring the data into the model.

Q2. Clean and transform the imported dataset: handle nulls, fix data types, and remove duplicate orders.

Sample Data:

OrderID	Sales	Profit
O1	4500	600
O1	4500	600
O4	null	150

Solution:

1. In Power Query, remove the duplicate O1 row using Remove Rows -> Remove Duplicates.

2. Replace or remove the null Sales value for O4 depending on whether it can be reasonably estimated or should be excluded.
3. Confirm OrderDate is Date type and Sales/Profit are Decimal/Whole Number types.

Q3. Build a data model connecting the Orders fact table to Product, Customer, and Date dimension tables.

Sample Data:

Fact/Dim	Key
Orders (fact)	ProductID, CustomerID, DateKey
Product / Customer / Date (dims)	ProductID / CustomerID / DateKey

Solution:

1. Load the three dimension tables alongside Orders.
2. In Model View, connect Orders to each dimension table on the shared key with One-to-Many cardinality (dimension = one side).
3. Arrange as a star schema with Orders at the centre.

Q4. Create at least five DAX measures relevant to the business questions (e.g. Total Sales, Total Profit, Profit Margin %, YoY Growth, Order Count).

Sample Data:

Measure	Formula
Total Sales	SUM(Orders[Sales])
Total Profit	SUM(Orders[Profit])
Profit Margin %	DIVIDE([Total Profit],[Total Sales])
Order Count	COUNTROWS(Orders)
YoY Sales Growth	DIVIDE([Total Sales]-CALCULATE([Total Sales],SAMEPERIODLASTYEAR('Date'[Date])),CALCULATE([Total Sales],SAMEPERIODLASTYEAR('Date'[Date])))

Solution:

1. Create each measure listed via Modeling -> New Measure, using the formulas shown.
2. Verify each measure returns a sensible value in a Card visual before using it in charts.

Q5. Design at least four visuals that answer specific business questions (e.g. 'Which region is most profitable?', 'What is the sales trend over time?').

Solution:

1. Bar chart: Total Profit by Region -> answers 'which region is most profitable'.
2. Line chart: Total Sales by Month/Year -> answers 'what is the sales trend over time'.
3. Pie/Donut chart: Sales share by Category -> answers 'which category drives the most revenue'.
4. Table/Matrix: Top 10 Products by Profit Margin % -> answers 'which products are most efficient, not just highest-selling'.

SET B

Q6. Add slicers/filters for Region, Category, and a Date range to make the dashboard interactive.

Solution:

1. Insert Slicer visuals for Region and Category, placed along the top or side of the dashboard.
2. Insert a Date slicer set to 'Between' style for flexible date-range selection.
3. Test that selecting values updates all visuals on the page consistently (cross-filtering).

Q7. Arrange all visuals into a polished dashboard layout, applying a consistent theme and aligned grid.

Solution:

1. Apply a built-in or custom theme via View -> Themes for consistent colors/fonts.
2. Group KPI cards at the top, trend/comparison charts in the middle, and detail tables/matrices toward the bottom.
3. Use Format -> Align and Distribute to keep all visuals neatly gridded.

Q8. Add a KPI visual with conditional formatting to flag regions or categories that are underperforming.

Sample Data:

Region	ProfitMargin%	Target%
North	22	20
East	-5	20

Solution:

1. Add a KPI visual (or Table with conditional formatting) comparing ProfitMargin% against Target%.
2. In a Table visual, select the ProfitMargin% column -> Conditional formatting -> Background color -> rule: red if below Target%, green if at/above.
3. East (-5%) would be flagged red as underperforming; North (22%) would show green.

Q9. Publish the completed mini-project dashboard to Power BI Service and share it with a defined set of stakeholders.

Solution:

1. Home tab -> Publish -> choose the appropriate workspace (e.g. 'Mini Project - Retail Analytics').
2. In the Service, pin key visuals to a Dashboard, or share the report directly with stakeholder email addresses set to Viewer access.
3. If applicable, configure Row-Level Security roles beforehand so each stakeholder sees only relevant data.

Q10. Prepare a one-page written summary of the key business insights derived from the dashboard.

Solution:

1. Summarize 3-5 headline findings, e.g. 'Electronics drives 45% of revenue but only 30% of profit', 'East region shows negative profit margin driven by high discounting', 'Q4 consistently outperforms other quarters by 20%+'.
2. Tie each insight back to a specific visual in the dashboard so stakeholders can verify it themselves.
3. End with 2-3 actionable recommendations, e.g. 'review discount policy in East region', 'increase Electronics inventory ahead of Q4'.

Assignment Evaluation

0. Not Done [] 1. Incomplete [] 2. Late Complete [] 3. Needs Improvement [] 4. Complete [] 5. Well Done []

Signature of the Instructor

Date: